KIDUS MESFIN TEFERI

Software Engineer

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SUMMARY

Software Engineer with 4+ years of experience across web, mobile, backend and applied AI, with a Bachelor's degree in Computer Science from HiLCoE School of Computer Science & Technology. Builds production systems end to end with React, Next.js, Node.js, TypeScript, Python, PostgreSQL and MongoDB, from real-time operations platforms to AI tooling, workflow automation and smart contracts. Trained with A2SV in advanced data structures and algorithms (Codeforces Expert, 1,000+ LeetCode problems), with experience leading engineering teams and teaching DSA. Passionate about leveraging technology to solve impactful real-world problems.

EXPERIENCE

Lead Systems & Platform Engineer, *Oz Kitchen* 04/2026
Remote

- Architected and maintained an end-to-end food production and delivery platform as the sole engineer, unifying customer ordering, kitchen operations, admin management, delivery logistics, and internal communication into one real-time system.
- Built scalable full-stack applications using React, Node.js, PostgreSQL, Supabase, Redis, and Upstash, improving platform performance, reliability, and maintainability across frontend, backend, and database layers.
- Implemented CI/CD pipelines with GitHub Actions and introduced multi-stage deployment workflows across development, staging, and production, reducing deployment risk and improving release consistency.
- Strengthened platform security by implementing authentication, role-based authorization, protected admin access, and controlled permissions across different user roles and operational workflows.
- Developed analytics, reporting, Telegram bot integrations, and customer communication automation to help stakeholders track performance, improve workflow efficiency, and make more data-driven business decisions.

Software Engineer — Full-Stack, Infrastructure & Automation, *Temaribet* 12/2025 – 05/2026
Education Technology Platform Remote

- Developed and maintained internal **moderator dashboards** and **backend infrastructure** supporting a large-scale education platform for Ethiopian high school students preparing for national exit examinations.
- Built scalable backend services using **Python (FastAPI)** with **Redis** for caching and performance optimization, and **RabbitMQ** for asynchronous task processing and event-driven workflows.
- Designed and implemented APIs and background workers to support content moderation, analytics, and system reliability.
- Developed cross-platform mobile features using **React Native** and modern web applications with **Next.js**, leveraging the **TypeScript** ecosystem for type safety and maintainability.

Founding Full-Stack Software Engineer, *Andro Solutions* 03/2025
Hybrid, Addis Ababa

- Led a cross-functional team to develop scalable web platforms for clients using **Next.js**, **Node.js**, and **MongoDB**.
- Optimized backend performance and implemented CI/CD pipelines, reducing deployment times by 40%.
- Mentored junior developers on best practices in software architecture and version control.

Head of Education, *A2SV* 01/2026 – 05/2026
Remote

- Led the design and delivery of **Data Structures and Algorithms (DSA)** training programs for software engineering students.
- Mentored students through competitive programming and real-world problem decomposition, improving consistency and performance.

Junior Software Developer, *The Idea Vault* 2022 – 01/2024
Kenya – Ethiopia

- Designed and maintained company websites and visual assets, improving user engagement by 25%.

PROJECTS

MaruCheck, *Open-Source Verification CLI for AI-Generated Code* 2026 – Present

- Built and maintain an MIT-licensed **TypeScript** CLI, published on npm, that checks every Git diff against owner-approved "Quality Contracts" so a change can't pass just because its tests were rewritten too.
- Ships a **GitHub Actions** pull-request gate with least-privilege permissions, job summaries, and uploaded evidence artifacts, mapping the release decision to the check status.
- Designed deterministic, explainable 0–100 risk scoring and requirement-linked verification planning that selects which Vitest, Playwright, Semgrep, and Gitleaks checks to run; works fully offline.
- Exposes an MCP server so Codex, Claude Code, and Cursor can run verification as tool calls, while contract approval stays human-owned; design decisions documented in 18 ADRs.

UnitCare, *Hospital EHR & Operations Platform (HiLCoE Senior Project)* 2026 – Present

- Building a phone-first hospital platform that unifies electronic health records with pharmacy, inventory, procurement, wards, theatre, finance and HR, so a lab order, its charge, the stock it consumes and its audit row are one transaction.
- Designed an event layer where clinical modules emit billing, stock and activity events, so a clinician's action reaches finance and inventory without any module knowing how the others work.
- Modelled 255 PostgreSQL tables across 36 domain files with Drizzle ORM and generated, checksum-tracked migrations; about 161k lines of strict TypeScript across 104 service modules and 35 API routes.
- Every server entry point enforces role- and scope-based permissions; an insert-only audit log covers every clinical read and write, with timed break-glass access that flags its audit rows and pages administrators.

Prospect AI, *AI-Powered Prospect Research & Outreach Platform* 04/2026 – 05/2026

Developed an AI-powered prospect research and outreach platform that extracts context from websites, GitHub profiles, notes, and screenshots, then generates personalized outreach messages and follow-up replies using OpenRouter text and vision models. Built with Next.js, React, TypeScript, Better Auth, Drizzle ORM, and PostgreSQL/Neon.

LendLoop, *Community Tool-Lending Platform* 2026

- Built a neighbourhood tool-lending app with Next.js 16, Convex, Clerk and TypeScript, modelling loans as a status-based state machine and writing events through a transactional outbox so none are lost or sent for rolled-back changes.
- Automated request notifications and daily overdue reminders in n8n with at-least-once delivery, idempotent write-backs, de-duplication keys, staged retries and a heartbeat alert if the schedule stalls.

Recruit Intel, *Football Recruitment Intelligence Platform* 2026

- Built a club-agnostic recruitment platform that profiles a squad, finds gaps against its tactical system and ranks market candidates to fill them; first configured for Manchester United.
- Trained player embeddings for similar-player search, clustering for role discovery, and quantile gradient-boosting models for defensive stats validated on held-out players with calibrated prediction ranges.

Orit Tej, *Membership Commerce Platform* 2026

- Built a Flutter app and Next.js API for a membership commerce business: member plans, bottle and case ordering, vendor onboarding, rotating pickup QR codes and notifications.

SKILLS

Languages
Python, TypeScript, JavaScript, Solidity, Go, C++, Java, C#, Dart, PHP

Databases & Data
PostgreSQL, MongoDB, Redis, Supabase, Convex, DuckDB, Firebase, Drizzle ORM

Applied AI & Automation
MCP, OpenRouter, embeddings, PyTorch, n8n, Telegram bots

Frameworks & Libraries
React, Next.js, Node.js, Express.js, FastAPI, Flask, Flutter, React Native, Gin

Cloud & DevOps
Docker, AWS, Vercel, GitHub Actions, CI/CD, RabbitMQ

Tools & Integrations
Git, Stripe, Clerk, Playwright, Figma, Postman

EDUCATION

Bachelor in Computer Science, *HiLCoE School of Computer Science & Technology* 09/2022 – 09/2026
Addis Ababa, Ethiopia

- **GPA:** 3.8 / 4.0.
- **Relevant Coursework:** Data Structures & Algorithms, System Design, Object-Oriented Programming, Software Engineering, Web Development.

Advanced Data Structures and Algorithms, *A2SV* 04/2025 – 01/2026
Kigali, Rwanda

Selected participant in a competitive 6-month mentorship program focused on advanced algorithms, problem-solving, and global tech exposure.

ACHIEVEMENTS

Codeforces, *Expert, peak rating 1726*
234 problems solved across 8 rated contests.

LeetCode, *1,001 problems solved*
156 hard, 573 medium and 272 easy problems.

LANGUAGE

- AMHARIC
- ENGLISH